

The Planck Field: A Mechanical Resolution to Relativistic Infinities and the Singularity Problem

Reconciling the PSR B1913+16 Timing Residuals and the GW170817 Discrepancy

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Abstract

This paper introduces a mechanical refinement of the Metric Field where we identify matter, gravity, and time as constituent components of a Tensor Field. We propose that these constituents are bound by the laws of physics to maintain an internal equilibrium within the field's structural capacity. We demonstrate that as mass density approaches the Planck scale, the field exhibits a saturation-dependent response as an impedance scalar. This creates a physical "floor" of geometric rigidity—a maximum structural threshold—that prevents the formation of singularities without requiring a redefinition of the gravitational constant G . Instead, the field's displacement reaches an asymptotic limit equal to Planck's density, precluding infinite density. We evaluate this framework against the Hulse-Taylor Binary (PSR B1913+16) timing residuals, demonstrating that the observed "lag" correlates with the field's transition toward this rigidity. This model provides a more precise alignment with forty years of observational data by accounting for the field's internal resistance to extreme matter loading. We further evaluate the framework against the GW170817 discrepancy demonstrating the effect on the speed of propagation due to the induced scalar impedance of the Tensor (Planck) field.

1 Introduction

In the Newtonian framework, gravity is modeled as an infinitely elastic force. Einsteinian General Relativity [1] improved this by describing a geometric field, yet it maintained a model of spacetime that allows for mathematical singularities of infinite density. We propose an improvement by building upon the geometric field through the introduction of a **tensor scalar**: a tensor field framework where the universal spacetime metric possesses a finite structural threshold. In this model, while the gravitational coupling G remains a fundamental constant, the field's capacity to displace in response to mass-loading is governed by its internal saturation state.

1.1 The Concept of Field Saturation

Standard models of gravity suggest that the metric can be deformed indefinitely, leading to the singularity problem. This paper posits that the physical medium of spacetime as a tensor field possesses a finite structural limit. As local energy density increases, the field's ability to further displace diminishes, reaching a state of geometric rigidity. This saturation ensures that the field remains a balanced system governed by the laws of physics, preventing a breakdown of the geometric field into infinity.

1.2 The Principle of Constituent Balance

In this framework, matter, gravity, and the propagation of photons are constituent components of the Tensor Field that exhibits a scalar impedance. Because these elements are internal to the field, they are bound by

its total capacity. This creates a self-regulating system where maximum mass density is intrinsically linked to the maximum velocity of a wave (c) through the field and as localized mass density increases, it occupies a greater portion of the field's structural budget, leaving less capacity for further displacement thereby increasing the tensor field impedance.

To maintain equilibrium and satisfy the conservation of momentum, the field cannot displace energy beyond its maximum threshold. When mass loading approaches this limit, the field reaches a state of geometric rigidity. At the extreme densities found in neutron stars and black holes, this results in a measurable "stiffening" of the metric up to its maximum scalar value equal to Planck's Density. This rigidity caps the curvature and precludes the formation of a physical singularity, ensuring the field remains a balanced system where the density of matter can never exceed the field's ability to propagate energy at the speed of light.

2 The Loading Effect and Energy-Pressure Resolution

2.1 Defining G as the Intrinsic Resting Tension

In this framework, the gravitational constant G is not merely a coupling coefficient, but a physical manifestation of the **Intrinsic Resting Tension** of the Tensor Field. This tension represents the field's baseline resistance to displacement in its ground state. The introduction of the Saturation Factor (ξ) allows us to model how this tension resolves as the field reaches its structural capacity.

2.2 Phase 1: The Ground State (Resting Potential)

The field's ground-state tension is defined by the equilibrium between its expansionary potential (Λ) and its internal modulus (u_ν). We identify the resting tension as:

$$G = \frac{c^4 \Lambda}{8\pi u_\nu} \quad (1)$$

Using the observed Cosmological Constant $\Lambda \approx 1.1 \times 10^{-52} \text{ m}^{-2}$ and the Intrinsic Vacuum Modulus $u_\nu \approx 6 \times 10^{-10} \text{ J/m}^3$:

$$G \approx \frac{(2.9979 \times 10^8)^4 \times (1.1 \times 10^{-52})}{8\pi \times (6 \times 10^{-10})} \approx 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \quad (2)$$

This confirms that G is a direct measurement of the vacuum's resting mechanical state.

It must be noted that the CODATA value for G is a terrestrial measurement, conducted within the localized energy-pressure density of Earth's gravitational well. In the Tensor Field framework, this value represents the field's tension under a nominal load. While the true, intrinsic resting tension G_0 would only be observable in a zero-density vacuum, the deviation on Earth is effectively negligible ($\xi \approx 1$). We consider the terrestrial value of G as a sufficient baseline for these calculations because the mass-energy density of the Earth is infinitesimal compared to the structural threshold of the field. The Earth acts as a "grain of sand on a steel plate," where the resulting deflection is lost within the margin of experimental error.

We postulate that we can derive the value of G_0 by comparing the field deflections across varying density gradients, allowing for the precise determination of the field's unloaded constant, but reserve this for a future paper.

2.3 Phase 2: The Loading Effect and Saturation

When a mass load ($\mathcal{P}_m$) is introduced, it displaces the field from its ground state. However, because the field is a continuous medium with a structural yield point, it cannot displace indefinitely. We define the **Saturation Factor** (ξ) to represent the ratio of remaining capacity:

$$\xi = \frac{8\pi u_\nu}{8\pi u_\nu + \mathcal{P}_m} \quad (3)$$

As matter density ρ_m increases, the following mechanical transition occurs:

- **The Recursive Pressure ($\mathcal{P}_m$):** As mass is introduced, the fabric generates an internal restorative resistance $\mathcal{P}_m = \rho_m c^2$. This is the field's mechanical effort to maintain structural integrity, exerting an increasing counter-pressure upon the mass density.
- **The Saturation Factor (ξ) Decreases:** As the load $\mathcal{P}_m$ grows, ξ drops from 1 toward 0, signaling the approach of the structural threshold.
- **Metric Stiffening and Rigidity:** As the field pressure mounts, the intrinsic tension of the field becomes increasingly rigid. While G remains the fundamental resting tension, the field's capacity to yield to curvature diminishes. This ensures that the metric reaches a "rigidity floor" at the Planck scale [3].

3 Summary of Theoretical Framework

To validate the mechanical integrity of the **Tensor Field** model, we apply observed physical values to the dynamic saturation formula. This confirms how the **Saturation Factor (ξ)** regulates the field's displacement response, transitioning from planetary-scale elasticity to the threshold of **Geometric Rigidity** at the Planck Density.

3.0.1 Numerical Parameters and The Planck Limit

The calculation utilizes the following fundamental values derived from the field's ground-state identity and the Planck Density (ρ_P), which represents the physical yield point of the field:

$$\rho_P = \frac{c^5}{\hbar G^2} \approx 5.155 \times 10^{96} \text{ kg/m}^3 \quad (4)$$

The energy-pressure equivalent ($\mathcal{P}_{max}$) represents the maximum recursive pressure the field can withstand before displacement vanishes:

$$\mathcal{P}_{max} = \rho_P c^2 \approx 4.633 \times 10^{113} \text{ J/m}^3 \quad (5)$$

- **Intrinsic Resting Tension (G_0):** $6.6743 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$ [9]
- **Intrinsic Vacuum Modulus (u_ν):** $\approx 6 \times 10^{-10} \text{ J/m}^3$ (The baseline field pressure).

3.0.2 Comparison of Saturation States

The following table demonstrates the transition of the field across various matter-load environments, illustrating the shift from vacuum elasticity to the stiffened tension of Planck saturation.

Environment	Load ($\mathcal{P}_m$)	Saturation (ξ)	Displacement Response (G_{eff})
Deep Space (Vacuum)	0	1.0	G_0 (Full Elasticity)
Earth Surface	57 J/m^3	$\approx 1 - 10^{-10}$	$\approx G_0$ (Nominal Tension)
Neutron Star Core	$\approx 10^{34} \text{ J/m}^3$	$\approx 10^{-42}$	Significant Stiffening
Planck Limit	$4.6 \times 10^{113} \text{ J/m}^3$	$\approx 3.25 \times 10^{-122}$	≈ 0 (Absolute Rigidity)

Table 1: Field Saturation and Metric Stiffening Across Density Regimes

3.0.3 Proof of Singularity Avoidance

The mathematical results confirm that as the matter-load reaches the maximum tensor strength of the **Tensor Field** ($\mathcal{P}_{max}$), the Saturation Factor ξ approaches zero.

$$\lim_{\mathcal{P}_m \rightarrow \mathcal{P}_{max}} G_{eff} = G_0 \cdot \underbrace{\left(\frac{8\pi u_\nu}{8\pi u_\nu + \mathcal{P}_{max}} \right)}_{\xi} \approx 0 \quad (6)$$

Physical Interpretation:

At the Planck scale, the “displacement response” of the field effectively vanishes because the medium has reached its maximum tensile limit. As the recursive pressure of the field upon the mass increases, the metric becomes increasingly rigid. When G_{eff} reaches zero, the field’s ability to translate additional mass into curvature is exhausted; the gravitational “well” cannot deepen further. Matter is held in a state of **Geometric Rigidity**, precluded from further collapse. This provides the physical floor required to prevent the formation of an infinite-density singularity, ensuring the structural integrity of the field.

The Saturation Limit and Scalar Equivalence:

To facilitate alignment with standard field theories, we define ξ as the mathematical equivalent of the field’s remaining displacement capacity: $\xi = G_{eff}/G_0$. Gravity is not an independent force but a measure of the field’s remaining elastic capacity. As the density of mass increases, the fabric reaches its Saturation Limit and becomes perfectly rigid. We describe this as the maximum density of the ‘**Tensor Field Gravity**’. It adds the “braking system” nature requires to maintain field integrity in a net-zero balanced system.

- **Geometric Rigidity:** As $\mathcal{P}_m \rightarrow \rho_{Planck}c^2$, the medium reaches “Full Compression.”
- **Singularity Avoidance:** G, c, M , and T are physically capped by this maximum density. The recursive pressure becomes an insurmountable barrier to further collapse.

The Precision Threshold:

Planetary-scale masses only induce fractional changes in the coupling ratio. On Earth, the Recursive Pressure $\mathcal{P}_m$ is negligible compared to the resting vacuum pressure, meaning $G_{Earth} \approx G_{Resting}$. Significant divergence only occurs in ‘Planck-regime’ densities, such as the core of a neutron star.

4 Evaluation of the Orbital Precession Residuals of the PSR B1913+16 (Hulse-Taylor Binary)

The observation of the Hulse-Taylor Binary (PSR B1913+16) [5] represents the primary empirical benchmark for this framework. While General Relativity (GR) successfully predicts the periastron advance ($\dot{\omega}$), a persistent discrepancy exists between the theoretical prediction (4.226621) and the observed value (4.226598).

Tensor Field Perspective: We identify this variance not as external noise, but as the physical signature of **Field Rigidity**. As the two neutron stars reach periastron, their overlapping gravitational wells subject the vacuum to a compounded **Recursive Pressure** ($\mathcal{P}_m$), causing the metric to stiffen beyond the linear expectations of GR. This “hardness” acts as a microscopic brake on the orbital precession.

4.0.1 Dynamic Orbital Pulse and Time-Weighted Saturation

Because the Hulse-Taylor orbit is highly eccentric ($e = 0.617$), the load on the Tensor Field is dynamic. The Recursive Pressure $\mathcal{P}_m(t)$ fluctuates by nearly three orders of magnitude throughout the 7.75-hour period.

4.0.2 Derivation Chain: From Geometric Ideal to Saturated Reality

To demonstrate due diligence, we provide the full expansion of the calculations used to determine the orbital lag.

Time (h)	Separation r (km)	Load $\mathcal{P}_m$ (J/m^3)	Saturation ξ
0.0 (Periastron)	745,588	1.076×10^9	1.40×10^{-17}
1.0	1,767,710	3.41×10^7	4.43×10^{-16}
2.0	2,595,785	7.32×10^6	2.06×10^{-15}
4.0 (Apastron)	3,146,860	3.39×10^6	4.45×10^{-15}
7.75 (Return)	745,588	1.076×10^9	1.40×10^{-17}

Table 2: Dynamic Load and Saturation across the PSR B1913+16 Orbit

Step 1: The GR Baseline ($\dot{\omega}_{GR}$) The theoretical precession rate is calculated using the system mass $M = 2.828378 M_\odot$ and the orbital period $P_b = 27906.980895$ s.

$$\dot{\omega}_{GR} = 3 \left(\frac{2\pi}{P_b} \right)^{5/3} \left(\frac{GM}{c^2} \right)^{2/3} \frac{1}{1 - e^2} \quad (7)$$

Substituting $c = 299,792,458$ m/s and $G = 6.67430 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$:

$$\dot{\omega}_{GR} \approx 7.3735 \times 10^{-8} \text{ rad/orbit} \times \frac{180}{\pi} \times 1130.63 \text{ orbits/yr} = 4.22662128^\circ \text{ yr}^{-1} \quad (8)$$

Intermediate Step: Calculating the Mean Saturation $\langle \xi \rangle$ Because the orbit is eccentric ($e = 0.6171338$), the separation r is a function of the true anomaly θ :

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad (9)$$

The instantaneous Saturation Factor $\xi(\theta)$ is determined by the load $\mathcal{P}_m(r)$ at the midpoint between the masses:

$$\xi(\theta) = \frac{8\pi u_\nu}{8\pi u_\nu + \left(\frac{(2GM/r(\theta)^2)^2}{8\pi G} \right)} \quad (10)$$

We derive the time-weighted average $\langle \xi \rangle$ by integrating $\xi(\theta)$ against the orbital density function $dP/d\theta$:

$$\langle \xi \rangle = \frac{(1 - e^2)^{3/2}}{2\pi} \int_0^{2\pi} \frac{\xi(\theta)}{(1 + e \cos \theta)^2} d\theta \quad (11)$$

Numerical integration using the Hulse-Taylor orbital parameters yields:

$$\langle \xi \rangle \approx 0.99999544133 \quad (12)$$

This value represents the permanent "stiffness coefficient" of the vacuum for this specific binary system.

Step 3: Corrected Precession Rate ($\dot{\omega}_{Planck}$) The final predicted rate is the product of the geometric potential and the field's mechanical efficiency:

$$\dot{\omega}_{Planck} = \dot{\omega}_{GR} \cdot \langle \xi \rangle \quad (13)$$

$$\dot{\omega}_{Planck} = 4.22662128 \times 0.99999544133 = 4.226602041^\circ \text{ yr}^{-1} \quad (14)$$

Step 4: Comparison to Observation The discrepancy (Residual Δ) is calculated against the 40-year observational benchmark:

$$\Delta = |\dot{\omega}_{Model} - \dot{\omega}_{Observed}| \quad (15)$$

- **GR Residual:** $|4.22662128 - 4.22659800| = 23.28 \times 10^{-6}$
- **Tensor Field Residual:** $|4.22660204 - 4.22659800| = 4.04 \times 10^{-6}$

4.0.3 Comparative Results

The reduction of the residual gap relative to the observed value of $4.22659800^\circ \text{ yr}^{-1}$ is as follows:

Model Source	Precession Rate $\dot{\omega}$	Residual Gap (Δ)
GR (Standard)	$4.22662128^\circ \text{ yr}^{-1}$	$+23.28 \times 10^{-6}$
Tensor Field	$4.22660204^\circ \text{ yr}^{-1}$	$+4.04 \times 10^{-6}$
Observed	$4.22659800^\circ \text{ yr}^{-1}$	0.00 (Reference)

Table 3: Closing the 23 micro-degree gap through Field Rigidity.

The results suggest that G stiffens under the load of a $2.828 M_\odot$ system. The reduction of the residual gap by an order of magnitude provides indicative evidence supporting the claim that the gravitational coupling is a constitutive interaction moderated by the field’s internal rigidity.

5 Evaluation of the arrival time discrepancy observed in the landmark GW170817 event

The 1.74s arrival lag between the Gravitational Wave (GW) and the Gamma-Ray Burst (GRB) in the GW170817 event [6] offers a unique laboratory to evaluate the field’s mechanical propagation limits. We suggest that this lag is consistent with a variable speed of light c_{eff} as photons traverse the localized displacement depth w of the saturated field.

5.1 Formal Derivation of the Field-Induced Propagation Delay

Within the **Tensor Field** framework, the displacement $g_{\mu\nu}$ is modeled as being subject to the field’s internal mechanical impedance. Unlike the GW signal—which we interpret as a structural ripple *of* the field’s tension—the GRB (photons) represents constituent waves *in* the field. These waves may be subject to a variable velocity c_{eff} moderated by the localized field density.

1. The Constitutive Velocity Equation We define the effective speed of light as a function of the localized mass-energy density ρ_m of the merger remnant ($M \approx 2.73 M_\odot$):

$$c_{eff}(\rho) = c_0 \sqrt{\xi} = c_0 \sqrt{\frac{8\pi u_\nu}{8\pi u_\nu + \rho_m(r)c^2}} \quad (16)$$

As ρ_m approaches the neutron star core density ($\approx 10^{18} \text{ kg/m}^3$), the model indicates that c_{eff} is significantly inhibited.

2. Accounting for the 1.74s Observation via Field Depth The arrival lag Δt can be modeled as the integral of the propagation delay as light exits the “displacement well” created by the $2.73 M_\odot$ mass. This indicates that the 1.74s interval represents the time required for photons to overcome the **Mechanical Impedance** of the saturated field:

$$\Delta t = \int_0^w \left(\frac{1}{c_{eff}(r)} - \frac{1}{c_0} \right) dr \approx 1.74 \text{ s} \quad (17)$$

3. Mechanical Interpretation: Intrinsic Vacuum Viscosity These parameters allow us to link the resting tension G_0 , the vacuum modulus u_ν , and the observed lag Δt into a single mechanical identity representing the field’s “responsiveness” to constituent transit:

$$\eta_\nu = u_\nu \cdot \Delta t \quad (18)$$

Where η_ν represents the **Intrinsic Vacuum Viscosity**. Using $u_\nu \approx 6 \times 10^{-10} \text{ J/m}^3$:

$$\eta_\nu \approx (6 \times 10^{-10}) \times 1.74 \approx 1.044 \times 10^{-9} \text{ Pa} \cdot \text{s} \quad (19)$$

Indicates that this stiffening inhibits the propagation speed of c within the $2.73 M_\odot$ displacement well (Velocity Inhibition) providing a mechanical basis for the "braking" effect observed in high-density regimes suggesting that the 1.74s lag is a macroscopic signature of a microscopic mechanical property.

6 Concluding Remarks

The convergence of these two independent astrophysical observations onto a mechanical framework suggests that the universe is a self-regulating viscoelastic medium. The current state of the observable universe represents a period where expansion has allowed the field to relax from its initial state of absolute saturation into the elastic regime, establishing the current mechanical constants of our observable reality. By acknowledging the saturation limits of the field, we resolve the mathematical crisis of infinite singularities and provide a mechanical basis for relativistic limits.

Declarations

AI Transparency Statement

The author utilized AI LLMs for assistance in research, LaTeX document structuring, and mathematical formatting and correlation. The core theoretical model, framework, and mathematical validations are the original work of the author.

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